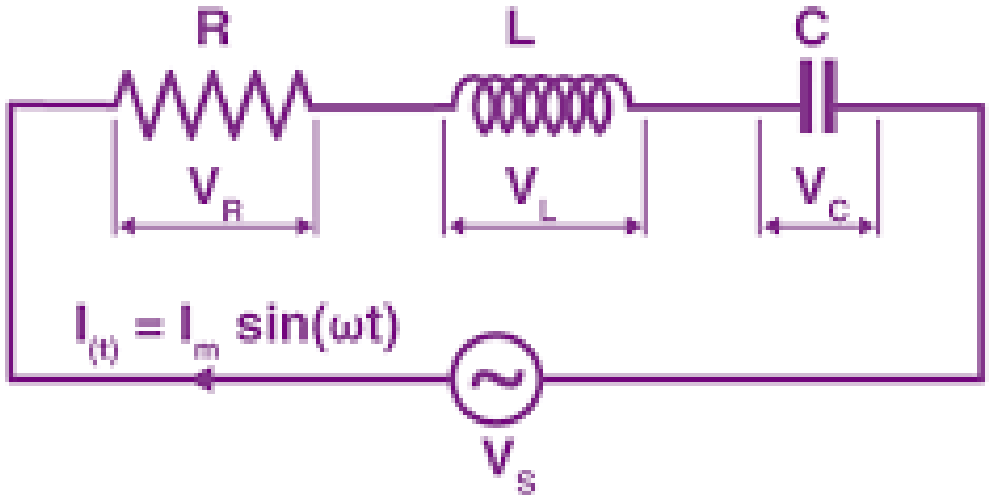
# AC Components




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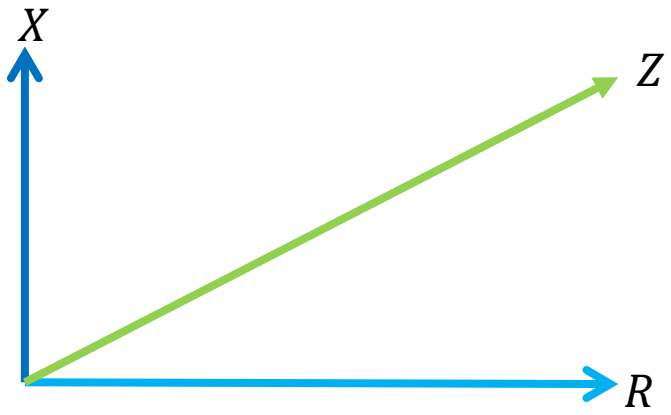
# DC vs AC circuit analysis

- In DC circuits we define ohms law as $V = IR$
- However, when working with AC circuits we use $V = IZ$
- Voltage and current are still the same however resistance is replaced with impedance ($Z$)



# Impedance

- Impedance is the $Z$ value in $V = IZ$
- It is the combination of two values:
  - Reactance ($X$) from capacitors and inductors
  - Resistance ($R$) from resistors

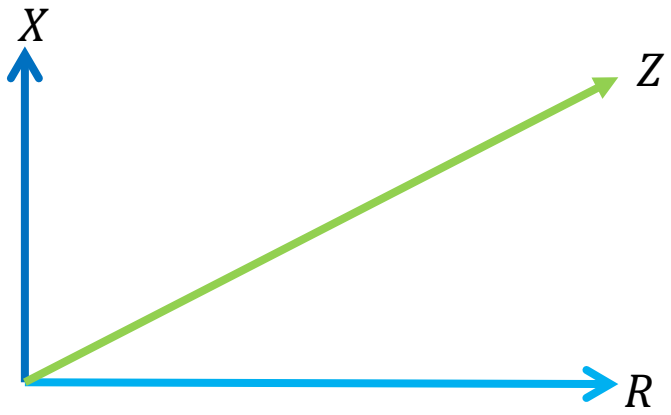


# Impedance

- When calculating impedance, we treat the value as a vector, we combine the $X$ and $R$ value using trigonometry
- This means polar impedance can be calculated using

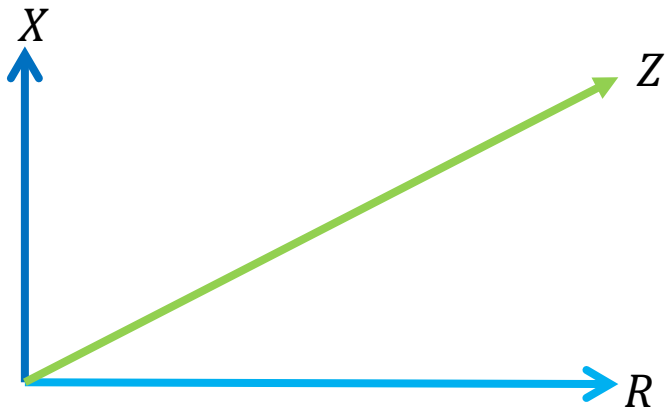
$$\boldsymbol{Z = \sqrt{R^2 + X^2}}$$

- We can then work out the angle of the polar impedance using $\boldsymbol{\tan^{-1}(\frac{X}{R})}$.
- Though when we do this make sure, we are in radian mode on our calculator



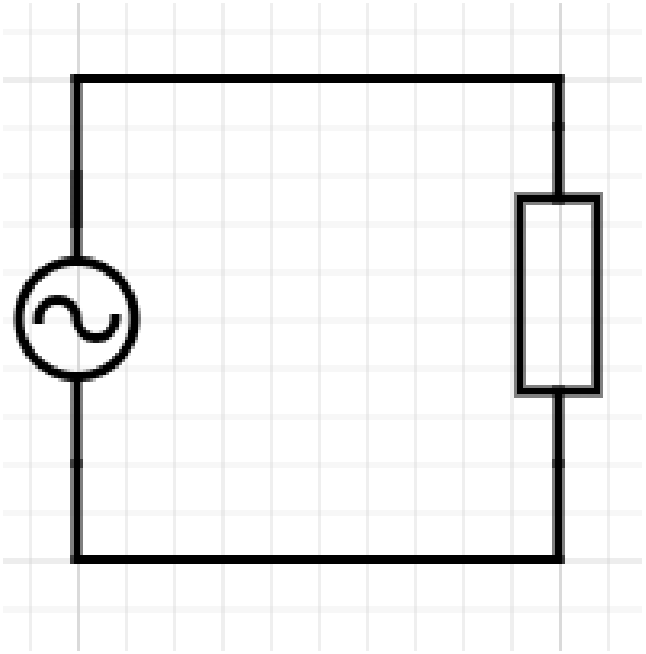
# Impedance

- Most of the time when doing complex AC circuit analysis, we leave our impedance in cartesian format making addition easier
- To add two impedances, we just combine the two $X$ components together and then the two $R$ components together
- So $Z_1 + Z_2 = (R_1 + R_2,\ X_1 + X_2)$



# Resistors

- When we have resistors in our circuit, they don't produce a phase shift, this means both voltage and current are aligned
- All they do is limit current reducing our peak current for the circuit
- If we have a purely resistive circuit, we can just use $V = IR$ to calculate our values the same as DC



V = IR @ instantaneous t

$V_{max} = I_{max} * R$

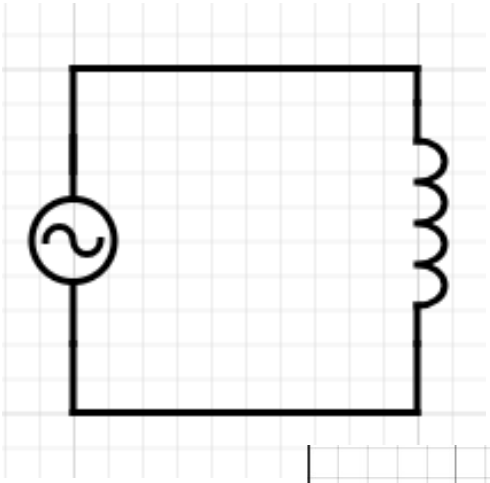
$V_{rms} = I_{rms} * R$

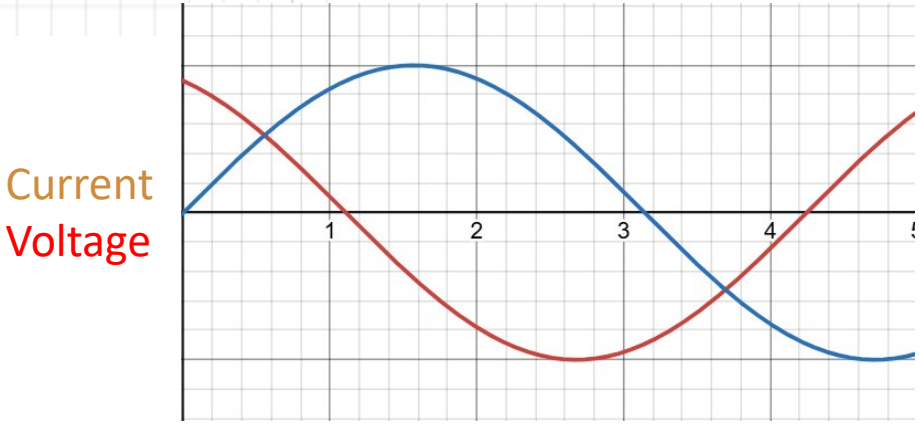
# Reactance

- Reactance is the $X$ value in our impedance calculations
- It incorporates all inductors and resistors in the circuit
- We measure Reactance in *ohms* $\Omega$
- Reactance is primarily used to tell us whether current or voltage is leading/lagging
- $+X \rightarrow$ voltage leads (inductive)
- $-X \rightarrow$ current leads (capacitive)

# Inductors

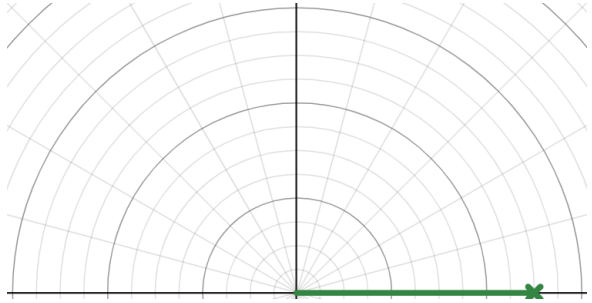
- Inductors cause the change in current to slow down (or phase shift)
- This is because the changing current through the inductor generates a changing magnetic field
- This changing magnetic field induces a voltage back into the inductor opposing the original current





# Inductive Reactance

- Inductors generate a **positive** reactance value which acts at $+90°$



- We can work out this reactance value using

$$\boldsymbol{X_L = \omega L} \textbf{ or } \boldsymbol{X_L = 2\pi f L}$$

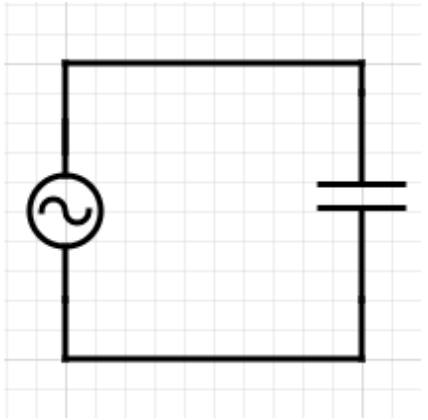
- This means for a purely inductive circuit we have

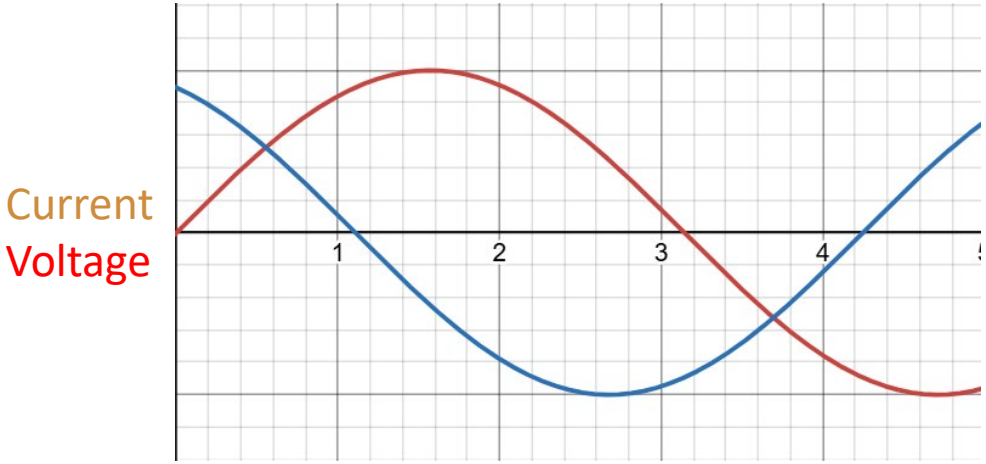
$$\boldsymbol{V = I * j\omega L} \textbf{ or } \boldsymbol{V = I * j2\pi f L}$$

Note we need to include the $j$ here to show we are treating this as an impedance with a direction not just a raw value, the $j$ isn't actually a variable

# Capacitors

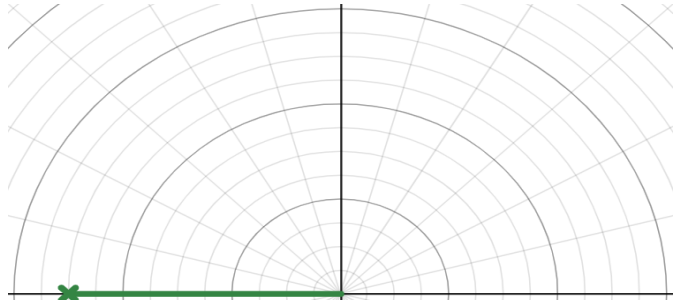
- Capacitors cause the change in voltage to slow down (or phase shift)
- This is because they take time to charge and discharge
- This causes the voltage to slip behind the current





# Capacitive Reactance

- Inductors generate a **negative** reactance value which acts at $-90°$



- We can work out this reactance value using

$$\boldsymbol{X_C = \frac{1}{\omega C}} \textbf{ or } \boldsymbol{X_C = \frac{1}{2\pi f C}}$$

- This means for a purely capacitive circuit we have

$$\boldsymbol{V = I * -\frac{1}{j\omega C}} \textbf{ or } \boldsymbol{V = I * -\frac{1}{j2\pi f C}}$$

Note we need to include the $j$ here to show we are treating this as an impedance with a direction not just a raw value, the $j$ isn't actually a variable

# Combining Inductors, Capacitors and Resistors (Series)

- For most series AC circuit questions, we can use a universal equation to give us an answer

$$\boldsymbol{Z = R + j(X_L - X_C)}$$

- This of course can then be put into the polar equation or left as it is in cartesian format, to convert to polar we use:

$$\boldsymbol{|Z| = \sqrt{R^2 + (X_L - X_c)^2}} \text{ and } \boldsymbol{tan^{-1}(\frac{(X_L - X_C)}{R})}.$$

# Combining Inductors, Capacitors and Resistors (Parallel)

- When we look at parallel components it gets slightly more difficult, we need to look at **admittance** ($\boldsymbol{Y}$) where:

$$\boldsymbol{Y = \frac{1}{Z}}$$

- This means we can work out the **overall impedance** of our parallel lines then find **their admittance**, to then **add them together:**

$$\boldsymbol{Y_{tot} = Y_1 + Y_2 + Y_3 + \cdots}$$

- We can then **convert back** to impedance by doing:

$$\boldsymbol{Z_{tot} = \frac{1}{Y_{tot}}}$$